Projections 1 - Multiple Choice Questions

1. What is the projection of vector v onto vector u given by?

A. $(u \bullet v)u / ||u||^2$

B. $(v \bullet u)u$

C. $(u \bullet v)v / ||v||^2$

D. $(v \bullet v)u / ||u||^2$

Answer: $(u \bullet v)u / ||u||^2$

2. If u is a unit vector, how can you simplify the projection formula?

A. $proj_u\, v = (v \bullet u)u$

B. $proj_u\, v = (u \bullet u)v$

C. $proj_u\, v = ||v||\, u$

D. $proj_u v = (u \bullet v)\, v$

Answer: A. $proj_u v = (v \bullet u)u$

3. What geometric relationship holds between $v - proj_u\, v$ and u?

A. They are parallel

B. They are perpendicular

C. They have equal magnitude

D. They point in opposite directions

Answer: They are perpendicular

4. If vectors u and v are perpendicular, what is $proj_u\ v$?

A. u

B. v

C. 0

D. u + v

Answer: 0 (just use the fact that $u \cdot v = 0$ if u and v are perpendicular.)

5. What happens to the projection $proj_u$ v if v is parallel to u?

A. It equals v

B. It equals u

C. It equals 0

D. It equals u - v

Answer: It equals v

6. Which expression correctly represents the magnitude of the projection $proj_u$v?

A. |(u • v)|

B. |v|

C. $|(u \cdot v)/||u||^2|$

D. |(u • v)/||u|||

Answer: |(u • v)/||u|||

7. If the angle between u and v is 90 degrees, what is true about their dot product u • v?

A. It equals 1

B. It equals -1

C. It equals 0

D. It equals ||u|| ||v||

Answer: It equals 0

8. When projecting v onto u, the resulting vector will always be:

A. Longer than v

B. Shorter than u

C. Parallel to u

D. Parallel to v

Answer: Parallel to u

9. Which vector expression correctly gives the component of v perpendicular to u?

A. v

B. $proj_u$ v

C. v - $proj_u$ v

D. $proj_u$ u

Answer: v - $proj_u$v

Important note: The midterm and the finals will NOT contain any questions on "Projections 2: on to a hyperplane."